

MI-27: The logarithmic commutator inequality with sharp constant one

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Abstract

For every pair of positive definite complex matrices A, B with $\operatorname{tr}(A + B) = 1$, we prove

$$\|[B, \log(A + B)]\|_1 \leq -\operatorname{tr}(A) \log \operatorname{tr}(A) - \operatorname{tr}(B) \log \operatorname{tr}(B).$$

The proof uses Frenkel's integral representation of quantum relative entropy, in its positive-part form given by Hirche and Tomamichel. The key estimate is that $\operatorname{tr}(\rho - \gamma e^{itH} \sigma e^{-itH})_+$ is $\|H\|_\infty$ -Lipschitz in t , uniformly for $\gamma \geq 1$ and for all matrix sizes. A positive integral of these estimates has total weight exactly the binary entropy. A strictly positive definite 2×2 family from the supplied findings pack proves that no coefficient smaller than one is possible.

1 Statement and conventions

The problem designated MI-27 in the linked collection asks specifically for coefficient one, rather than merely a dimension-independent coefficient [1]. Its historical formulation and a coefficient-two bound are recorded in [2, 3]. The following settles the stated matrix inequality.

Theorem 1.1 (MI-27, with the optimal universal coefficient). *Let $n \geq 1$, and let $A, B \in \mathbb{C}^{n \times n}$ be positive definite with $\operatorname{tr}(A + B) = 1$. Put $a = \operatorname{tr} A$ and $b = \operatorname{tr} B$. Then*

$$\boxed{\|[B, \log(A + B)]\|_1 \leq h(b)}, \quad h(b) = -a \log a - b \log b, \quad a + b = 1. \quad (1.1)$$

The coefficient one is best possible even when $n = 2$ and both matrices are strictly positive definite.

All logarithms are natural. The commutator is $[X, Y] = XY - YX$; $\|X\|_1 = \operatorname{tr} \sqrt{X^* X}$ is the full trace norm, not one half of it; and $\|X\|_\infty$ is the operator norm. A density matrix is a positive semidefinite matrix of trace one. For Hermitian X , X_+ is its positive part. Write

$$S(\rho) = -\operatorname{tr} \rho \log \rho, \quad D(\rho \| \sigma) = \operatorname{tr} \rho (\log \rho - \log \sigma),$$

with the usual support convention for relative entropy and $0 \log 0 = 0$. Only positive definite matrices are needed in the application to Theorem 1.1.

Proof dependency. The only non-elementary input is the published relative-entropy identity in Section 3. Everything used after that identity is proved below. The argument does not assume that A and B commute, that either has rank one, or that the projection reduction in the supplied pack has already been solved.

2 A uniform positive-part estimate

Lemma 2.1 (Commuting with an effect). *If $X \geq 0$ and $0 \leq Q \leq I$, then*

$$\|[X, Q]\|_1 \leq \operatorname{tr} X. \quad (2.1)$$

Proof. The Hermitian matrix $R = 2Q - I$ is a contraction. Consequently, the triangle inequality and the trace-norm product inequality give

$$\|[X, Q]\|_1 = \frac{1}{2} \|XR - RX\|_1 \leq \frac{1}{2} (\|XR\|_1 + \|RX\|_1) \leq \|X\|_1 \|R\|_\infty \leq \operatorname{tr} X.$$

□

For density matrices ρ, σ and $\gamma \geq 1$, define

$$\mathbf{E}_\gamma(\rho \|\sigma) = \operatorname{tr}(\rho - \gamma\sigma)_+. \quad (2.2)$$

This is often called the quantum hockey-stick divergence.

Lemma 2.2 (Uniform unitary-orbit Lipschitz bound). *Let ρ, σ be density matrices, let $H = H^*$, and set $U_t = e^{itH}$ and $\sigma_t = U_t \sigma U_t^*$. For every $\gamma \geq 1$ and every $s, t \in \mathbb{R}$,*

$$|\mathbf{E}_\gamma(\rho \|\sigma_t) - \mathbf{E}_\gamma(\rho \|\sigma_s)| \leq |t - s| \|H\|_\infty, \quad (2.3)$$

$$|\mathbf{E}_\gamma(\sigma_t \|\rho) - \mathbf{E}_\gamma(\sigma_s \|\rho)| \leq |t - s| \|H\|_\infty. \quad (2.4)$$

Proof. For every Hermitian M , diagonalization shows that

$$\operatorname{tr} M_+ = \max_{0 \leq Q \leq I} \operatorname{tr} QM. \quad (2.5)$$

In particular $|\operatorname{tr} M_+ - \operatorname{tr} N_+| \leq \|M - N\|_1$. Thus $f(t) = \operatorname{tr} M(t)_+$, where $M(t) = \rho - \gamma\sigma_t$, is locally Lipschitz, hence absolutely continuous on compact intervals and differentiable almost everywhere.

At a point of differentiability, fix the spectral projection $Q_t = \mathbf{1}_{(0, \infty)}(M(t))$. It maximizes (2.5), and it commutes with $M(t)$. We have

$$f(t + u) \geq \operatorname{tr} Q_t M(t + u) = f(t) + u \operatorname{tr} Q_t M'(t) + o(|u|).$$

The positive and negative choices of u , together with differentiability of f at t , imply $f'(t) = \operatorname{tr} Q_t M'(t)$. No differentiability of Q_t is being assumed.

Since $[\rho - \gamma\sigma_t, Q_t] = 0$, we have $[\rho, Q_t] = \gamma[\sigma_t, Q_t]$. Therefore

$$\begin{aligned} f'(t) &= -i\gamma \operatorname{tr} Q_t [H, \sigma_t] = -i\gamma \operatorname{tr} H [\sigma_t, Q_t] \\ &= -i \operatorname{tr} H [\rho, Q_t]. \end{aligned} \quad (2.6)$$

Trace-norm duality and Lemma 2.1 now yield

$$|f'(t)| \leq \|H\|_\infty \|[\rho, Q_t]\|_1 \leq \|H\|_\infty.$$

Integrating this almost-everywhere bound for the absolutely continuous function f proves (2.3).

For (2.4), use unitary invariance:

$$\mathbf{E}_\gamma(\sigma_t \|\rho) = \mathbf{E}_\gamma(\sigma \|\rho U_t^*),$$

and apply the first result with generator $-H$. □

Remark 2.3. The cancellation in (2.6) removes the apparently large factor γ . A generic perturbation bound on the second argument would keep that factor and is not sufficient here. Eigenvalue crossings at zero cause no gap: the scalar function is absolutely continuous, and its derivative is used only almost everywhere.

3 The relative-entropy representation and its kernels

Frenkel's Theorem 6 [4], in the equivalent form of Hirche–Tomamichel, Corollary 2.3, equation (2.22) [5], states that for density matrices ρ, σ ,

$$D(\rho\|\sigma) = \int_1^\infty \left\{ \frac{E_\gamma(\rho\|\sigma)}{\gamma} + \frac{E_\gamma(\sigma\|\rho)}{\gamma^2} \right\} d\gamma. \quad (3.1)$$

This is the ordinary Umegaki relative entropy even when the matrices do not commute. It is not a replacement of relative entropy by a different quantum divergence.

For reference, Frenkel's original state-normalized form is

$$D(\rho\|\sigma) = \int_{-\infty}^\infty \frac{\operatorname{tr}((1-v)\rho + v\sigma)_-}{|v|(v-1)^2} dv.$$

The integrand vanishes for $0 \leq v \leq 1$. In the part $v > 1$, the substitution $\gamma = v/(v-1)$ gives the first integral in (3.1). In the part $v < 0$, the substitution $\gamma = (v-1)/v$ gives the second. Here $X_- = (-X)_+$.

Lemma 3.1 (Skew relative-entropy kernel). *For $0 < \alpha < 1$, put $\beta = 1 - \alpha$. Then*

$$\begin{aligned} D(\rho\|\alpha\rho + \beta\sigma) \\ = \int_1^\infty \left\{ \frac{\beta^2}{\gamma(\beta + \alpha\gamma)^2} E_\gamma(\rho\|\sigma) + \frac{\beta^2}{(\alpha + \beta\gamma)^2} E_\gamma(\sigma\|\rho) \right\} d\gamma. \end{aligned} \quad (3.2)$$

Proof. Let $M = \alpha\rho + \beta\sigma$. For $1 \leq v < 1/\alpha$,

$$E_v(\rho\|M) = (1 - \alpha v) E_{\beta v/(1-\alpha v)}(\rho\|\sigma),$$

and for $v \geq 1/\alpha$ the positive part is zero. For all $v \geq 1$,

$$E_v(M\|\rho) = \beta E_{(v-\alpha)/\beta}(\sigma\|\rho).$$

Apply (3.1) to (ρ, M) . In its first integral make the change of variables

$$\gamma = \frac{\beta v}{1 - \alpha v}, \quad v = \frac{\gamma}{\beta + \alpha\gamma}, \quad dv = \frac{\beta}{(\beta + \alpha\gamma)^2} d\gamma.$$

In its second integral put $v = \alpha + \beta\gamma$. These give precisely the two coefficients in (3.2). Every integrand is nonnegative; the substitutions are valid also for improper integrals. $\square$

For $a, b > 0$ with $a + b = 1$, define the weighted entropy difference

$$\begin{aligned} \chi_{a,b}(\rho, \sigma) &:= S(a\rho + b\sigma) - aS(\rho) - bS(\sigma) \\ &= aD(\rho\|a\rho + b\sigma) + bD(\sigma\|a\rho + b\sigma). \end{aligned} \quad (3.3)$$

The second equality follows by expanding the traces.

Proposition 3.2 (A positive kernel of mass $h(b)$). *For density matrices ρ, σ ,*

$$\boxed{\chi_{a,b}(\rho, \sigma) = ab \int_1^\infty \left\{ \frac{E_\gamma(\rho\|\sigma)}{\gamma(b + a\gamma)} + \frac{E_\gamma(\sigma\|\rho)}{\gamma(a + b\gamma)} \right\} d\gamma.} \quad (3.4)$$

The two nonnegative scalar kernels have respective masses

$$ab \int_1^\infty \frac{d\gamma}{\gamma(b + a\gamma)} = -a \log a, \quad (3.5)$$

$$ab \int_1^\infty \frac{d\gamma}{\gamma(a + b\gamma)} = -b \log b. \quad (3.6)$$

Proof. Use Lemma 3.1 first with $(\alpha, \rho, \sigma) = (a, \rho, \sigma)$, then with $(\alpha, \rho, \sigma) = (b, \sigma, \rho)$, and insert the results in (3.3). The coefficient of $E_\gamma(\rho||\sigma)$ simplifies as follows:

$$\frac{ab^2}{\gamma(b+a\gamma)^2} + \frac{ba^2}{(b+a\gamma)^2} = \frac{ab}{\gamma(b+a\gamma)}.$$

The other coefficient follows by exchanging a and b .

For the first mass, an antiderivative is $a \log(\gamma/(b+a\gamma))$. At $\gamma = 1$ it equals zero, and at infinity it tends to $-a \log a$. The other mass is identical with a, b exchanged. Also $0 \leq E_\gamma \leq 1$ for density matrices and $\gamma \geq 1$, so (3.4) is absolutely convergent. $\square$

4 Proof of the upper bound

Proposition 4.1 (Finite-time entropy estimate). *Let $A, B > 0$ with $\text{tr } A = a$, $\text{tr } B = b$, $a + b = 1$, and let $H = H^*$. For $S_t = A + e^{itH} B e^{-itH}$,*

$$|S(S_t) - S(S_0)| \leq |t| \|H\|_\infty h(b) \quad \text{for all } t \in \mathbb{R}. \quad (4.1)$$

Proof. Set $\rho = A/a$, $\sigma = B/b$, and $\sigma_t = e^{itH} \sigma e^{-itH}$. Unitary invariance of entropy gives

$$S(S_t) - S(S_0) = \chi_{a,b}(\rho, \sigma_t) - \chi_{a,b}(\rho, \sigma).$$

Apply the two bounds of Lemma 2.2 to the two integrands in (3.4). Integrate the resulting finite-difference bounds and use (3.5)–(3.6):

$$\begin{aligned} |S(S_t) - S(S_0)| &\leq |t| \|H\|_\infty ab \int_1^\infty \left\{ \frac{1}{\gamma(b+a\gamma)} + \frac{1}{\gamma(a+b\gamma)} \right\} d\gamma \\ &= |t| \|H\|_\infty (-a \log a - b \log b). \end{aligned}$$

The kernels are integrable and independent of t . In particular, *no differentiation under an improper integral is needed.* $\square$

Proof of the inequality in Theorem 1.1. Put $S = A + B$. The path S_t in Proposition 4.1 is positive definite near zero (in fact for every t), so its entropy is differentiable. The derivative of the trace of a matrix function gives

$$\left. \frac{d}{dt} S(S_t) \right|_{t=0} = -\text{tr}(S'_0(\log S + I)) = -i \text{tr}[H, B] \log S = -i \text{tr } H[B, \log S],$$

where $S'_0 = i[H, B]$ and $\text{tr } S'_0 = 0$. Equivalently, the trace derivative formula follows by diagonalizing S and summing the diagonal terms of the Fréchet derivative of $-x \log x$.

Divide (4.1) by $|t|$ and let $t \rightarrow 0$. With the Hermitian matrix $K = -i[B, \log S]$, this yields

$$|\text{tr } HK| \leq \|H\|_\infty h(b) \quad (H = H^*). \quad (4.2)$$

Choose $H = \text{sgn}(K)$, using zero on $\ker K$. Then $\|H\|_\infty \leq 1$ and $\text{tr } HK = \text{tr } |K| = \|K\|_1 = \|[B, \log S]\|_1$. Inequality (4.2) is exactly (1.1). $\square$

5 Sharpness within the strictly positive definite class

The following family is retained, with attribution, from the MI-27 findings in the user's supplied pack [6]; its properties are verified here rather than assumed. For $0 < t < 1/2$, set

$$S_t = \begin{pmatrix} 1-t & 0 \\ 0 & t \end{pmatrix}, \quad d_t = (1-2t^2)t(1-t), \quad B_t = t^2 S_t + d_t \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad A_t = S_t - B_t. \quad (5.1)$$

Let $u_t = (\sqrt{t}, \sqrt{1-t})^\top$ and $P_t = u_t u_t^*$. This is a rank-one orthogonal projection, and

$$S_t^{1/2} P_t S_t^{1/2} = t(1-t) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Consequently,

$$B_t = S_t^{1/2} (t^2 I + (1-2t^2) P_t) S_t^{1/2}.$$

The middle matrix has eigenvalues t^2 and $1-t^2$, both strictly between zero and one. Thus $A_t, B_t > 0$, and $\text{tr}(A_t + B_t) = \text{tr} S_t = 1$. As an additional algebraic check,

$$\det A_t = \det B_t = t^3(1-t)(1-t^2) > 0.$$

Writing $b_t = \text{tr} B_t$, direct calculation gives

$$b_t = t^2 + 2d_t, \quad \|[B_t, \log S_t]\|_1 = 2d_t \log \frac{1-t}{t}. \quad (5.2)$$

Indeed, the commutator is real skew-symmetric with off-diagonal magnitude $d_t \log((1-t)/t)$, so its two singular values both equal that magnitude.

As $t \downarrow 0$, $d_t/t \rightarrow 1$ and $b_t/(2t) \rightarrow 1$, whence

$$\frac{2d_t \log((1-t)/t)}{2t \log(1/t)} \rightarrow 1.$$

Moreover, $h(x) = x \log(1/x) + x + O(x^2)$ as $x \downarrow 0$. Since $\log(1/b_t)/\log(1/t) \rightarrow 1$, it follows that

$$\frac{h(b_t)}{2t \log(1/t)} \rightarrow 1.$$

Therefore

$$\lim_{t \downarrow 0} \frac{\|[B_t, \log(A_t + B_t)]\|_1}{h(\text{tr} B_t)} = 1. \quad (5.3)$$

For every $c < 1$, the ratio in (5.3) exceeds c for sufficiently small positive t . Thus no smaller universal coefficient is valid under the original strict hypotheses. This completes the sharpness part of Theorem 1.1.

A The corresponding skew-divergence improvement

Define the normalized skew divergence, for $0 < \alpha < 1$, by

$$\text{SD}_\alpha(\rho \parallel \sigma) = \frac{D(\rho \parallel \alpha \rho + (1-\alpha)\sigma)}{-\log \alpha}.$$

The kernels in (3.2) have masses

$$\begin{aligned} \int_1^\infty \frac{\beta^2}{\gamma(\beta + \alpha\gamma)^2} d\gamma &= -\log \alpha - \beta, \\ \int_1^\infty \frac{\beta^2}{(\alpha + \beta\gamma)^2} d\gamma &= \beta, \quad \beta = 1 - \alpha. \end{aligned}$$

For the first integral one may use the antiderivative $\log \gamma - \log(\beta + \alpha\gamma) + \beta/(\beta + \alpha\gamma)$; for the second use $-\beta/(\alpha + \beta\gamma)$. Thus Lemma 2.2 also proves

$$|\mathrm{SD}_\alpha(\rho\|U_t\sigma U_t^*) - \mathrm{SD}_\alpha(\rho\|\sigma)| \leq |t| \|H\|_\infty. \quad (\text{A.1})$$

The same holds when the first argument is rotated, by joint unitary invariance. This supplies the coefficient-one estimate in the skew-divergence approach to mixing; compare the coefficient-two unitary perturbation estimate in [3, Section 8]. Indeed, expanding

$$\chi_{a,b}(\rho, \sigma) = -a \log a \mathrm{SD}_a(\rho\|\sigma) - b \log b \mathrm{SD}_b(\sigma\|\rho)$$

and using (A.1) gives a second presentation of Proposition 4.1.

B Connection to the pack's projection reduction

The supplied pack [6] reduced the target to the assertion

$$\left\| [S^{1/2} P S^{1/2}, \log S] \right\|_1 \leq h(\mathrm{tr} SP), \quad S > 0, \quad \mathrm{tr} S = 1, \quad (\text{B.1})$$

for every orthogonal projection P , with no rank restriction. Theorem 1.1 now proves this assertion: put

$$K_\varepsilon = \varepsilon I + (1 - 2\varepsilon)P, \quad B_\varepsilon = S^{1/2} K_\varepsilon S^{1/2}, \quad A_\varepsilon = S - B_\varepsilon,$$

apply the theorem for $0 < \varepsilon < 1/2$, and let $\varepsilon \downarrow 0$. The sum S and its logarithm stay fixed, and all other quantities are continuous. For $P = 0$ or $P = I$, both sides of (B.1) are zero, with $h(0) = h(1) = 0$.

For completeness, the converse reduction is also valid. If $0 \leq K \leq I$, write its eigenvalues as $k_1 \geq \dots \geq k_n \geq 0$, put $k_{n+1} = 0$, and let P_j project onto the first j eigenvectors. Then

$$K = \sum_{j=1}^n (k_j - k_{j+1}) P_j + (1 - k_1) 0$$

is a convex combination of projections. For the corresponding weights θ_j , put $B_j = S^{1/2} P_j S^{1/2}$ and $b_j = \mathrm{tr} B_j$. Linearity, trace-norm convexity, (B.1), and concavity of h give

$$\left\| [S^{1/2} K S^{1/2}, \log S] \right\|_1 \leq \sum_j \theta_j h(b_j) \leq h\left(\sum_j \theta_j b_j\right).$$

Take $K = S^{-1/2} B S^{-1/2}$ to recover the original assertion. Importantly, the b_j are allowed to vary; no assertion is made about extreme points in a fixed-trace slice. The upper-bound proof in the main text does not require this reduction.

C Verification, scope, and provenance

The accompanying `verification/verify.py` checks the rational kernel identities and the sharpness-family determinants symbolically. It also compares the relative-entropy and weighted-entropy integrals with direct matrix evaluations, exercises a zero-eigenvalue crossing, checks trace-norm duality and entropy derivatives, and evaluates the strictly positive definite sharpness family at high precision. The quadrature is split at generalized eigenvalues of the matrix pencils to handle the positive-part kinks. The complete numerical output, software versions, random seed, and tolerances are included.

These computations are consistency checks, not the proof of a universal inequality. The analytic proof is Sections 1–5, using the published identity (3.1). The report is not a proof-assistant formalization and has not been independently refereed. No claim of historical publication priority is made. The source and exact input pack provenance are recorded in `provenance.md`; the sharpness family and projection reduction are explicitly credited to that pack.

References

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- [6] User-supplied findings archive, *OpenProblemsInNLA matrix proof checkpoint, 2026-09-12*, MI-27/`result.md`, Theorems 1–2. The archive supplies the projection equivalence and the strictly positive definite sharpness family, but explicitly leaves the universal upper bound unproved. An unchanged copy of that MI-27 result is included in `provenance/input_MI27_result.md`.